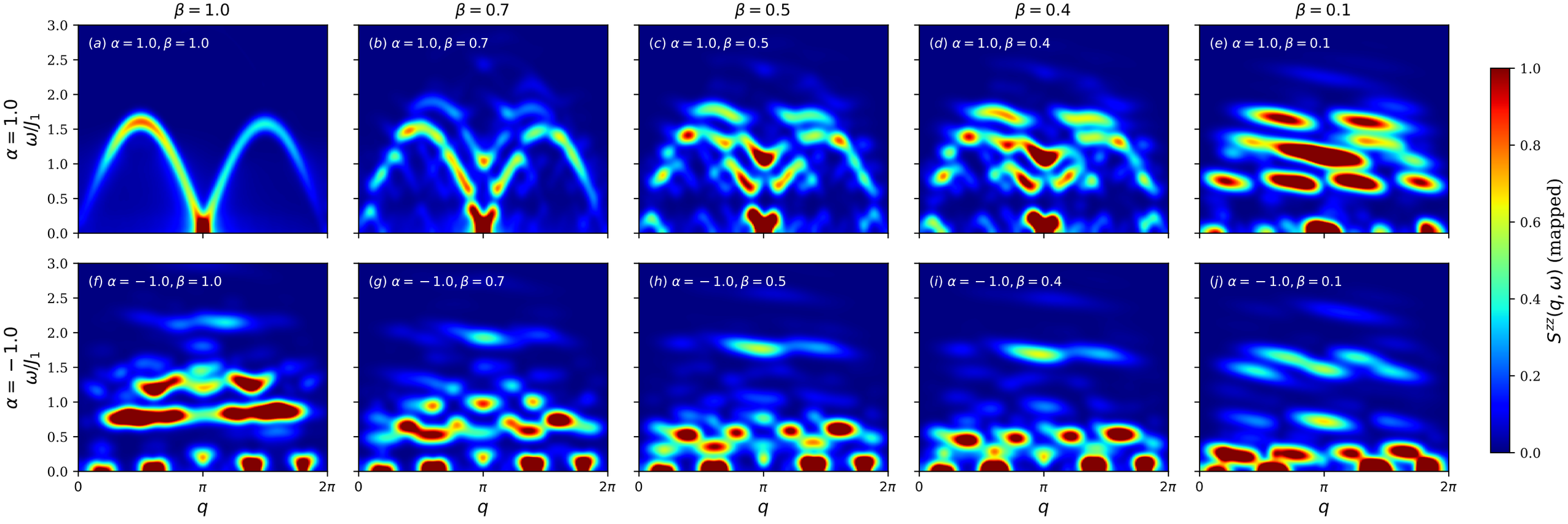

$\beta = 1.0$
$\beta = 0.7$
$\beta = 0.5$
$\beta = 0.4$
$\beta = 0.1$
(a) $\alpha = 1.0, \beta = 1.0$
(b) $\alpha = 1.0, \beta = 0.7$
(c) $\alpha = 1.0, \beta = 0.5$
(d) $\alpha = 1.0, \beta = 0.4$
(e) $\alpha = 1.0, \beta = 0.1$
(f) $\alpha = -1.0, \beta = 1.0$
(g) $\alpha = -1.0, \beta = 0.7$
(h) $\alpha = -1.0, \beta = 0.5$
(i) $\alpha = -1.0, \beta = 0.4$
(j) $\alpha = -1.0, \beta = 0.1$
$\alpha = 1.0$
$\alpha = -1.0$
ω/J_1
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
π
2π
q
$S^{zz}(q, \omega)$ (mapped)
0.0
0.2
0.4
0.6
0.8
1.0